

WARRN Full Knowledge Base

Overview

WARRN stands for Wildlife/Animal Rescue and Reporting Network. It is a smart rescue coordination platform that connects the public with volunteers, NGOs, and administrators to help injured, abandoned, or endangered animals quickly using digital tools and AI automation.

Core Modules

1. Incident Reporting: Users upload photos, location, and notes about injured animals.
2. AI Species Detection: Gemini AI analyzes uploaded images to identify animal species.
3. Live Video Monitoring: OpenCV processes live camera feeds to detect accidents or distress.
4. Real-Time Coordination: Responders receive instant dashboard and email alerts.
5. Adoption Board: Rescued animals can be listed for adoption.
6. Analytics Dashboard: Tracks rescue activity and outcomes.

Workflow

Reporter submits case → System stores report → Nearby responders notified → Volunteer claims case → NGO escalation if needed → Animal rescued/treated → Case resolved → AI generates clinical summary.

User Roles

Public Reporter, Volunteer, NGO Staff, Adopter, Administrator.

Technology Stack

Frontend: HTML/CSS/JavaScript/Jinja2 templates.

Backend: Flask, SQLAlchemy, Flask-Login, Flask-SocketIO.

Database: SQLite/PostgreSQL.

AI: Google Gemini, OpenCV.

Automation: Flask-Mail, ReportLab.

AI Features

Species identification from images, veterinary-style case narratives, smart summaries, chatbot support, future predictive analytics for hotspot zones.

RAG Chatbot Use Cases

How do I report an injured animal?
How does species detection work?
How are volunteers notified?
How to adopt an animal?
What happened in recent rescue cases?
What NGOs are available nearby?

Future Scope

Mobile app, multilingual chatbot, emergency hotline integration, drone surveillance, map heatmaps, hospital integrations, donation portal.